Activity 3

Differentiate total station from other levelling instruments and submit a report.

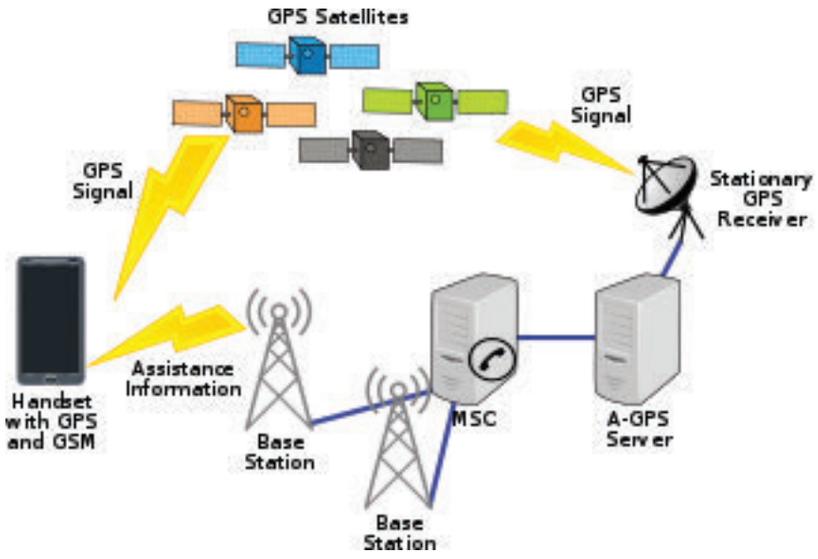
3.4.3 Global Positioning System (GPS)

The Global Positioning System (GPS) is a satellite based navigation and surveying system. It is adopted for the determination of precise position and time using radio signals from satellites.

GPS is a U.S space based radio navigation system freely available to all civilian user in all weather conditions, day and night anywhere in the world.

Now a days, GPS is finding its way into cars, boats, planes, construction equipments, farm machinery, cell phones and even in laptop computers.





Fundamentals of GPS

GPS basically comprised of 3 segments

1. **Control segment:** Control segment consist of one master control station located at Colorado (USA) and five monitor stations located throughout the world.
2. **Space segment:** It consist of group of 24 satellites in 6 orbital planes. 4 satellites in each plane 20.200 km altitude and 55˚ inclination.
3. **User segment:** It consist of antennas, receiver, processor, display and a regulated DC power supply.

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